ADRIN JOSEPH

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PROFESSIONAL SUMMARY

Aspiring AI Engineer and Software Developer pursuing an online BS degree at IIT Madras and undertaking software development training at Brototype. Interested in Python development, Artificial Intelligence, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and backend engineering. Developing independent projects focused on advanced retrieval systems and reliable, cost-controlled LLM infrastructure. Interested in building scalable and efficient AI applications.

EDUCATION

Indian Institute of Technology Madras (IIT Madras) *2025 – 2029*
Online BS Degree Programme Expected Graduation: 2029

PROFESSIONAL DEVELOPMENT

Brototype *May 2026 – Present*
Software Development Training

TECHNICAL SKILLS

Programming Languages: Python, SQL Fundamentals

Databases: PostgreSQL, Relational Database Concepts

AI and LLM Technologies: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Embeddings, Text Chunking, Information Retrieval, Prompt Engineering

Backend Development: REST API Fundamentals, API Architecture, HTTP, JSON, Backend Development Concepts

Cloud and DevOps (Currently Learning): Amazon Web Services (AWS), Docker, Linux, CI/CD, GitHub Actions, Cloud Deployment

LLM Infrastructure (Currently Exploring): Rate Limiting, Retry Mechanisms, Fault Tolerance, Provider Failover, Token Usage Tracking, LLM Cost Management

Developer Tools: Git, GitHub, Git Bash

TECHNICAL PROJECTS

Advanced-RAG *Ongoing*

Advanced Retrieval-Augmented Generation System
Tech Stack: Python, PostgreSQL, LLMs

- Designing a modular Retrieval-Augmented Generation (RAG) system for building knowledge-grounded applications using LLMs.
- Exploring document ingestion, text preprocessing, intelligent chunking, embedding generation, and retrieval pipeline architecture.
- Studying retrieval strategies, context construction, PostgreSQL integration, and retrieval evaluation to understand the reliability and performance of RAG systems.

ModelGate *Ongoing*

Multi-Provider LLM Gateway
Tech Stack: Python, API Architecture, LLM Infrastructure

- Designing a reliable, cost-controlled LLM gateway to manage requests across multiple upstream model providers.
- Exploring a provider-independent request handling architecture to integrate multiple LLM services through a unified interface.
- Designing resilience mechanisms involving retry strategies, rate limiting, and provider failover.
- Exploring token usage tracking and budget management mechanisms to monitor and control LLM API expenditure.